PAULSTRETCH · EXPLAINED WITH PICTURES

How Paulstretch works

Making a sound last much longer, without changing its pitch: what the module does, one step at a time.

Paulstretch (“Paul's Extreme Sound Stretch”) was invented by Nasca Octavian Paul.

For the Paulstretch and Paulstretch Heads modules, plugin version 2.1.0 · 7 November 2026

Paulstretch can turn a two-second chord into a minute-long, slowly breathing cloud of sound. It does this without the sound dropping in pitch, which is what normally happens when you slow something down.

This guide shows how, using pictures of real sounds going through the same steps as the module. You don't need any background: each picture comes with a short explanation and a few things to look for.

In this guide

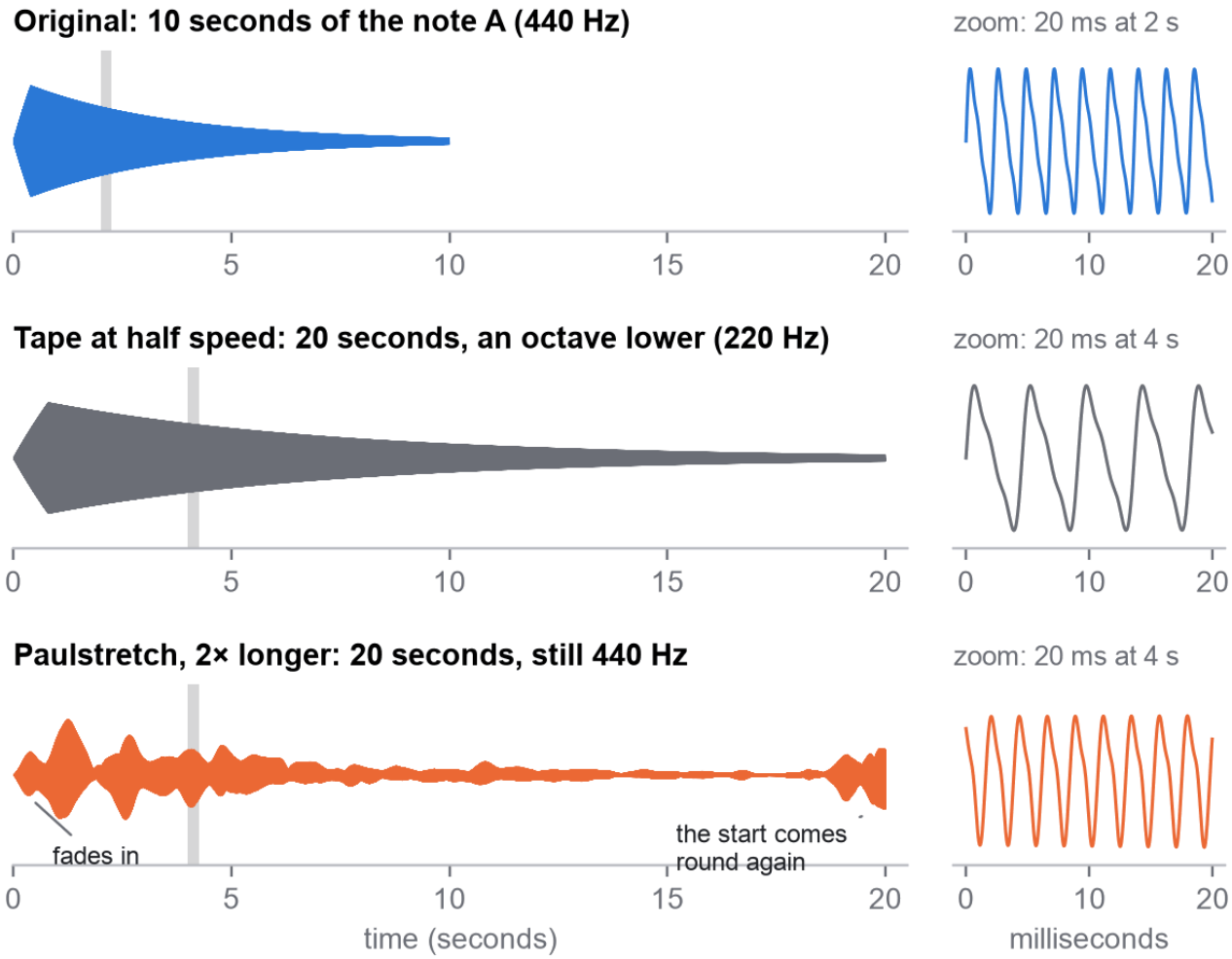
- Why slowing down usually changes the pitch
- Two ways to look at a sound
- The five steps: slice, list the pitches, give each a random phase shift, overlap, move on slowly
- What comes out
- Why sharp sounds become soft
- Small and large windows
- Pitch, freeze and reverse
- Several voices on one sound, and how they share the work

1

Why slowing down usually changes the pitch

Sound is air moving quickly back and forth. How fast it moves sets the **pitch**: the note A, for example, moves back and forth 440 times a second, written 440 Hz. A drawing of that movement over time is called a **waveform**.

Play a tape at half speed and every movement takes twice as long. The sound lasts twice as long, but it also moves back and forth half as often, so it drops an octave. Paulstretch makes the sound last longer while keeping how fast it moves, so the pitch stays where it was.



Below, a 10-second note is stretched to 20 seconds both ways. The wide strips show the whole sound; the small ones on the right zoom in on 20 milliseconds of each, at the same moment of the note.

What to look for

- Top: the note lasts 10 seconds.
- Both stretched versions last 20 seconds. Count the waves in the zooms: about nine, then four and a half (the tape, an octave lower), then nine again (Paulstretch, the same note).
- Paulstretch fades in over its first moment, and its loudness wobbles: the random phase shifts (step 3) does that to a single held note. It still dies away slowly, like the original.
- At the very end the note's start comes round again: the module plays the sound in a loop.

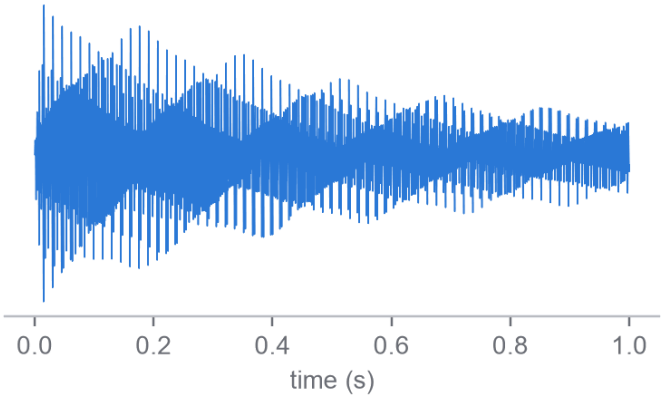
2

Two ways to look at a sound

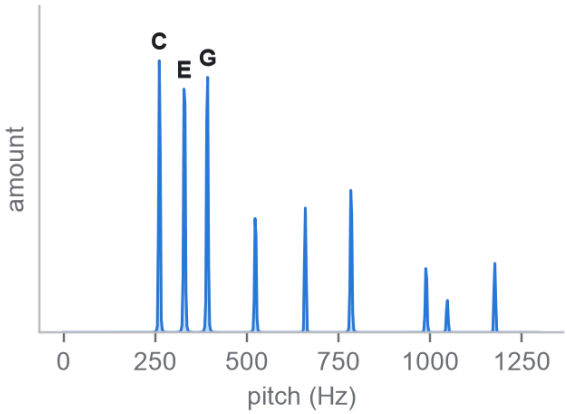
A waveform shows how a sound moves over time. A **spectrum** shows the same sound another way: as a list of the pitches it contains and how much of each. Think of it as the sound's list of ingredients.

Below is a C major chord, three notes played together. Its waveform is a dense tangle. Its spectrum is easy to read: one peak for each note, plus smaller peaks higher up. Those smaller peaks, called harmonics, are what make a plucked string sound different from a flute playing the same note.

Waveform: how it moves over time



Spectrum: which pitches it contains



What to look for

- The three tall peaks on the right are the notes C, E and G.
- Peaks further right are higher pitches. The height of a peak is how much of that pitch there is.

Later pictures use a **spectrogram**: a spectrum for every moment, laid side by side. Time runs left to right, pitch runs bottom to top, and darker means louder. A held note is a horizontal line.

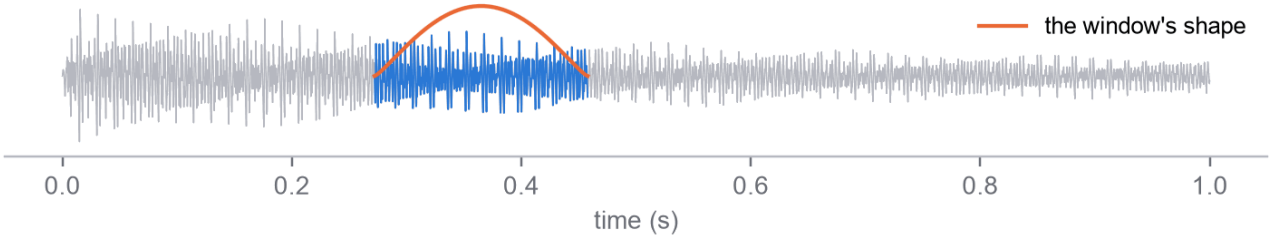
3

Step 1: take a short slice

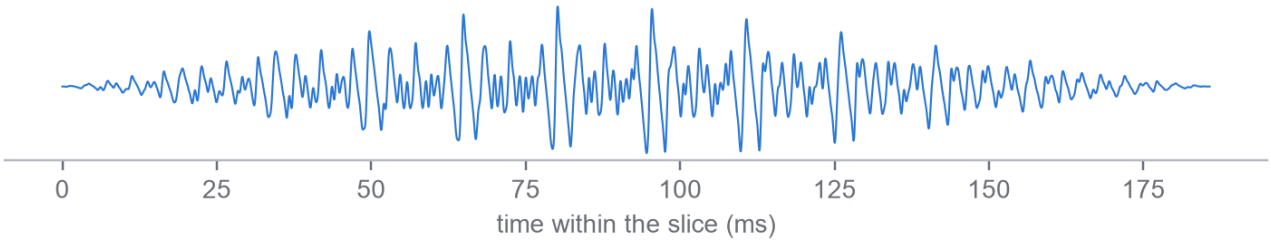
Paulstretch works on short slices of the sound, one after another. The length of a slice is the **window**, the WINDOW knob on the module. Here it is about a fifth of a second.

Each slice is faded in and out, following the orange curve, so its edges are silent. Without the fade, every slice would start and stop abruptly, and you would hear a click each time.

1 · Take a short slice of the sound (the window)



The slice, faded in and out by the window



What to look for

- The shaded part is the slice. The orange curve is the fade.
- In the lower picture, the slice starts and ends at zero: no sudden edges.

4

Steps 2 and 3: list the pitches, then shift their phases at random

Next, the module works out the slice's spectrum: which pitches are in it and how much of each. Every pitch in that list has two parts: an **amount** (how loud) and a **phase**: where its wave is in its up-and-down cycle at the start of the slice, which sets exactly when its waves rise and fall.

Paulstretch keeps every amount exactly as it was, and gives every pitch a random phase shift: its wave is slid along by a random part of one cycle. Then it rebuilds the slice from that list. The waveform comes out looking completely different, but the ingredients are the same, so it sounds like the same moment: the same notes, at the same loudness.

This is the heart of Paulstretch. The random phases remove everything that happens *within* the slice, such as the exact moment a note starts. What's left is pure colour, which is why the result sounds so smooth.

Below, the slice is taken apart into its notes, one colour each, with their higher harmonics in grey. The rebuilt slice uses the same coloured waves, each with its own random phase shift.



What to look for

- Each coloured wave keeps its pitch and size; only where its waves rise and fall has moved.
- Added together, the two sides look nothing alike: the waves no longer line up the same way.
- Bottom: the two spectra lie on top of each other. The shading shows where each coloured wave comes from.

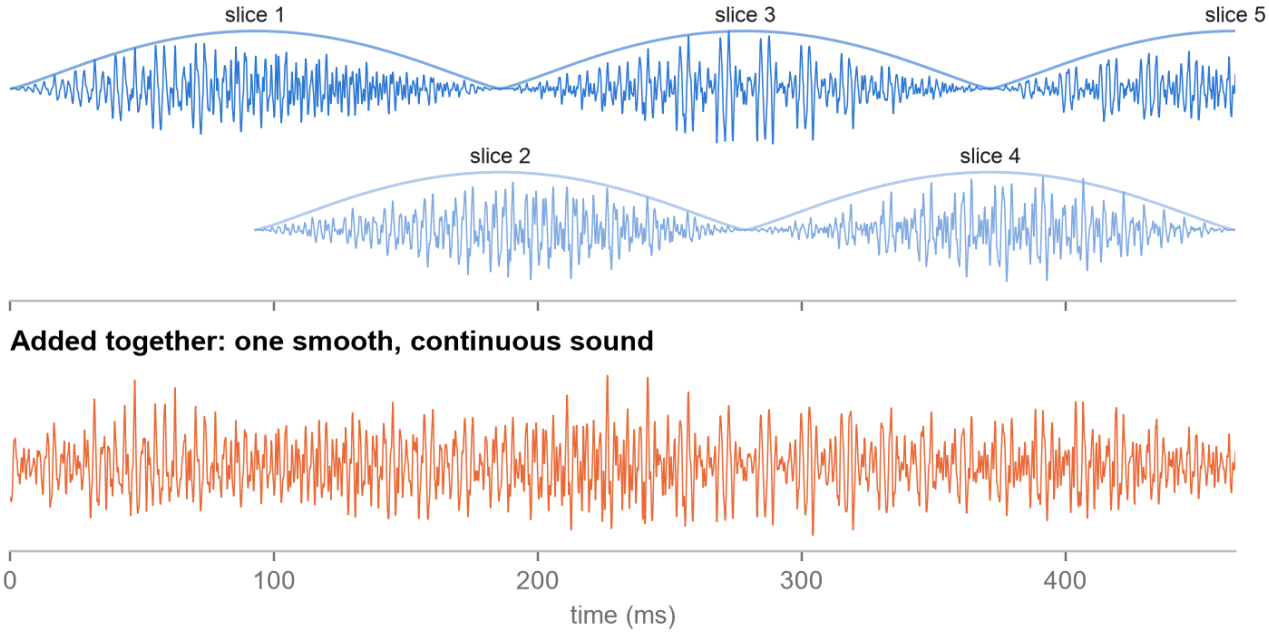
A simplification: the spectrum really splits a slice into about two thousand closely spaced pitches, and each gets its own random phase. The C above is a small cluster of them around 261 Hz.

5

Step 4: overlap the slices

A single rebuilt slice fades in and out. To make one continuous sound, the slices overlap: each starts halfway through the one before. Where one slice fades out, the next fades in, and added together they give an even, unbroken sound.

2 · Each rebuilt slice starts half a slice after the one before



What to look for

• Top: rebuilt slices, drawn on two rows so you can see each one. Each starts half a slice after the previous one.

• Bottom: their sum. The fades fill each other's gaps, so the level stays even.

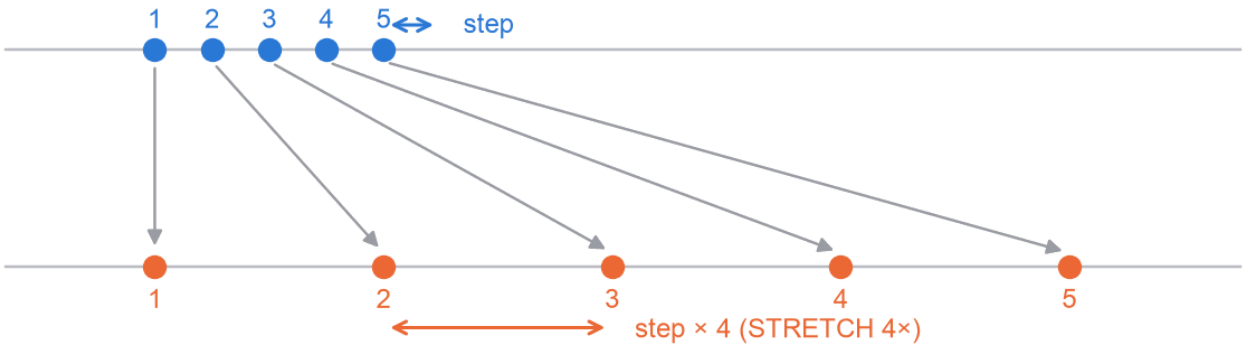
6

Step 5: move through the input slowly

Here is where the stretching happens. On the output side, a new slice starts every half window, always the same distance apart. On the input side, each slice is read from only a little further on than the one before.

With STRETCH at 4×, the input moves forward a quarter of the distance the output does. So it takes the output four times as long to get through the input: one second of input becomes four seconds of output. At 8× it becomes eight seconds; at 1000×, about seventeen minutes.

Input: the middle of each slice it reads, a small step apart



Output: the middle of each rebuilt slice, a big step apart

What to look for

- Top row: where each slice is taken from the input. Small steps.
- Bottom row: where each rebuilt slice is placed in the output. Steps four times as big.
- Nothing is repeated or skipped: the input is simply walked through more slowly.

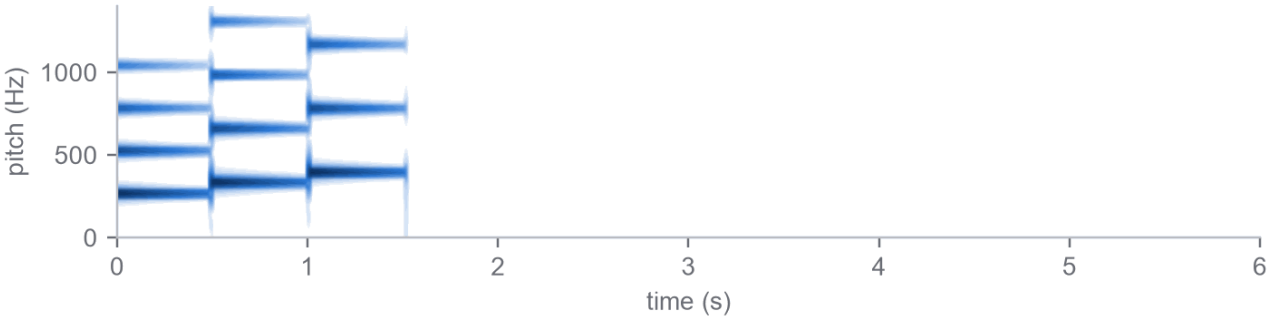
7

What comes out

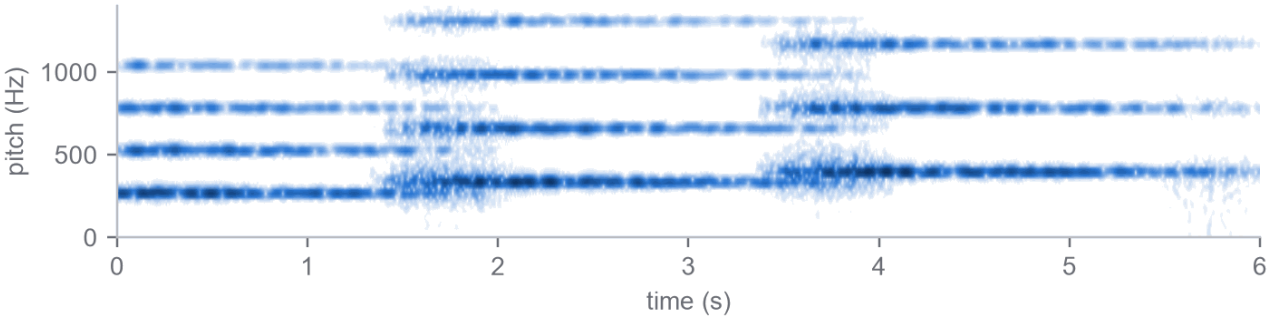
Put the five steps together and this is the result. The input is three plucked notes, C, E and G, one after another. Stretched 4×, the output plays the same three notes, in the same order and at the same pitches, over four times as long.

Two things change besides the length. Each note no longer has a clear start; it fades in. And where one note changes to the next, the two blend, because a slice can span both notes at once.

Input: three plucked notes, C, E, G, in 1.5 s



Output at 4× stretch: the same notes, 6 s long, smooth and blended



What to look for

- The horizontal lines sit at the same heights in both pictures: the pitches haven't changed.
- The output lasts four times as long, and the changes between notes are soft and gradual.

8

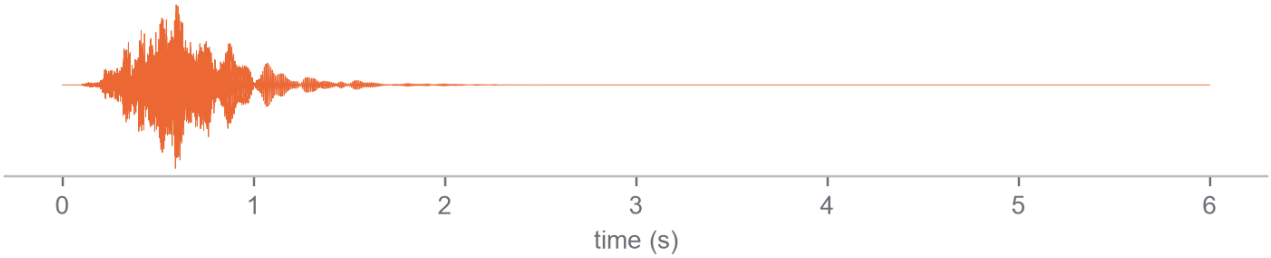
Why sharp sounds become soft

A drum hit is almost all attack: a sudden start, then a fade. The start is sudden because, at that instant, all its pitches line up. In step 3, Paulstretch shifts every pitch's phase at random, so they no longer line up, and the hit loses its edge and becomes a soft swell of the same sound, rising and fading over the length of a few slices.

Input: a drum hit, sudden then fading



Output at 4× stretch: no sudden start, a soft swell and fade



What to look for

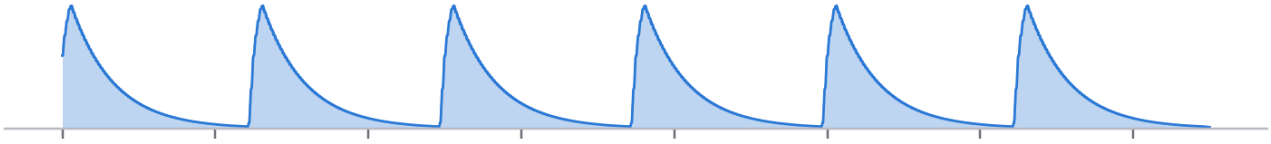
- Top: the hit starts in an instant.
- Bottom: the stretched version has no instant start. It rises and fades smoothly, and it lasts longer.

9

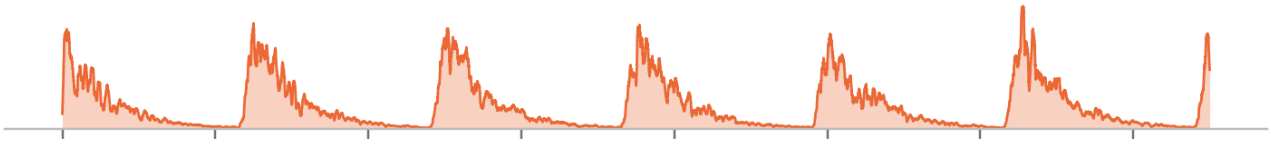
Small and large windows

The window size sets how much of the sound each slice covers, and so how much detail survives the random phase shifts. A small window takes many short slices: separate events still come through, and the sound is grainier. A large window takes long slices that each cover several events: they merge into one smooth sound.

Input: six short notes, 1.5 s



Small window, stretched 4×: the six notes still come through



Large window, stretched 4×: one smooth, even sound



What to look for

- Top: six separate short notes.
- Middle, small window: six bumps are still there, rougher.
- Bottom, large window: the bumps are gone. One steady, even sound.

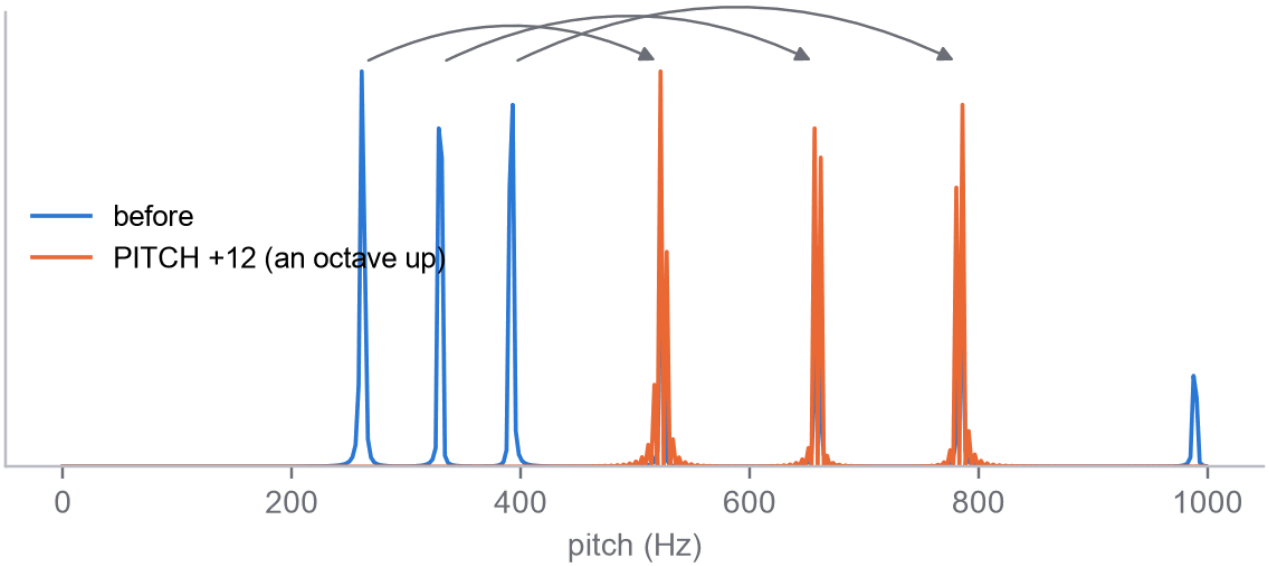
On the module, WINDOW goes from 1k (1,024 samples, about 21 milliseconds) to 64k (65,536 samples, about 1.4 seconds). The default, 16k, is about a third of a second.

10

PITCH: moving every ingredient

Because Paulstretch already has each slice as a list of pitches, changing the pitch is simple: before rebuilding, it moves every ingredient up or down by the same musical interval. An octave up doubles every pitch. The speed of the stretch isn't affected.

Every ingredient moves to twice its pitch



What to look for

• Each blue peak moves to twice its pitch, shown in orange. The C, E and G of the chord become the C, E and G an octave higher.

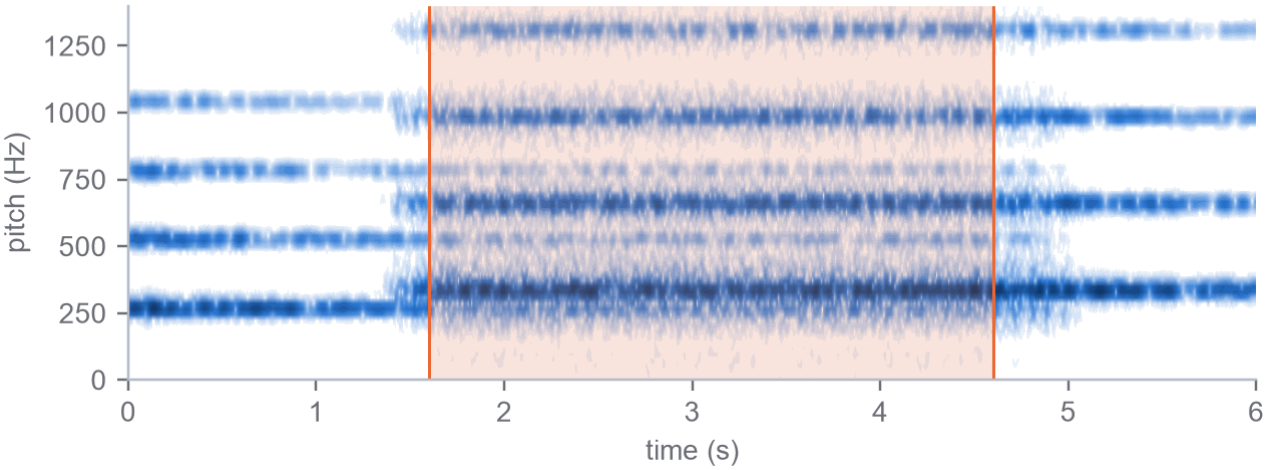
• The highest blue peak, near 1,000 Hz, has no orange partner on this chart: it moved to near 2,000 Hz, off the right edge.

11

FREEZE: holding one moment

FREEZE stops step 5: the module keeps reading slices from the same place. The output doesn't loop or go silent, though. Every new slice gets fresh random phases in step 3, so it is the same moment, rebuilt again and again. The result is a steady, slowly shimmering drone.

Stretched melody, with FREEZE on in the shaded part: one moment, held



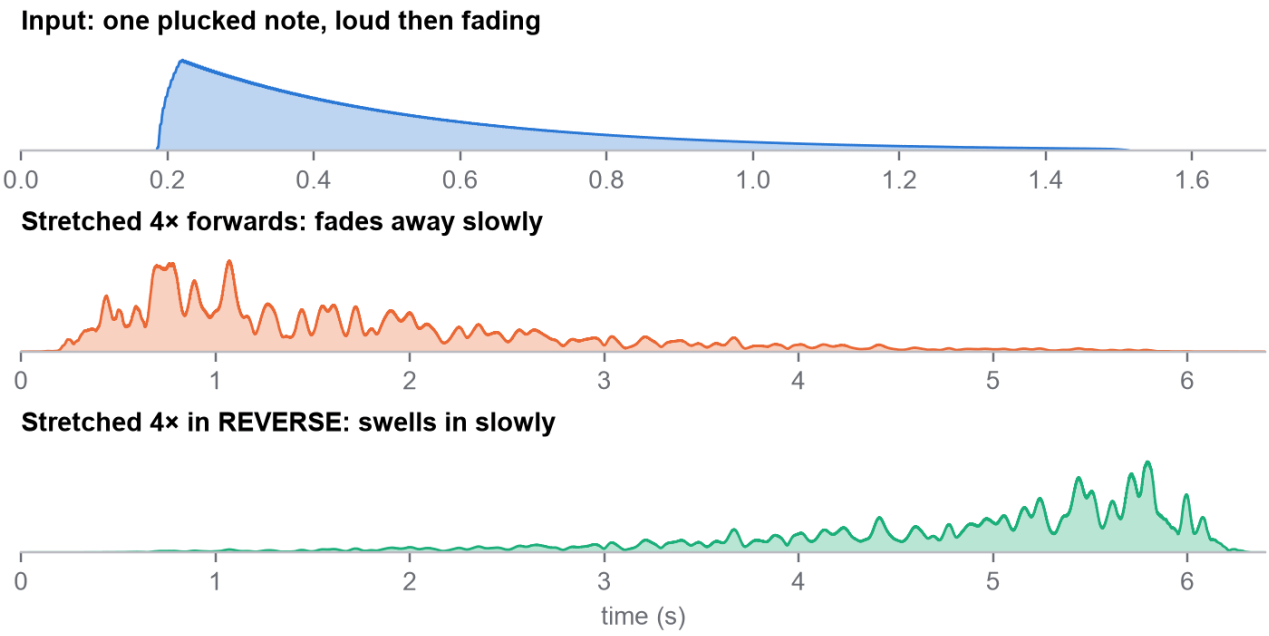
What to look for

- Inside the shaded part, the lines stay flat: the sound holds still.
- The held slice happens to span the end of the C and the start of the E, so you hear a bit of both.
- When FREEZE is released, the melody carries on from where it stopped.

12

REVERSE: walking backwards

REVERSE runs step 5 backwards: each slice is read from a little earlier than the one before. Inside a slice there is no forwards or backwards, because its phases are random anyway. So what changes is the shape over time: a note that faded away now swells in.



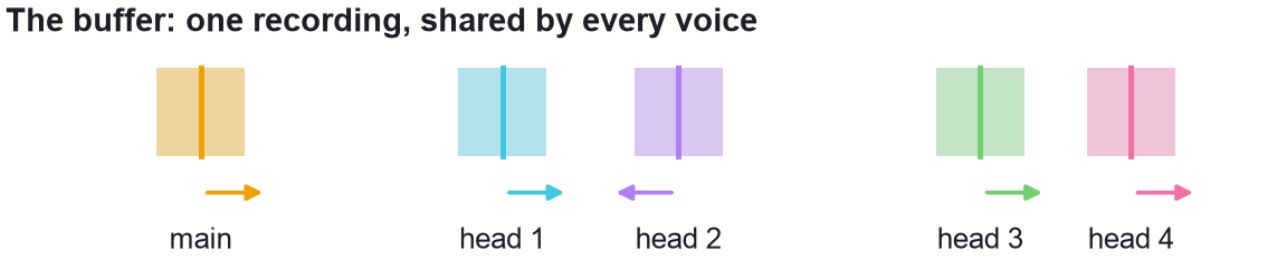
What to look for

- Top: a plucked note, loud at the start and fading.
- Middle, forwards: it fades away, slowly.
- Bottom, in reverse: the same note grows from silence to its loudest at the end.

13

Several voices on one sound

The Paulstretch Heads expander adds more voices, called heads, that read the same recording. Each head goes through the same five steps, but from its own place, at its own stretch, in its own direction and at its own pitch. Together they can make chords, or layers moving at different speeds.



What to look for

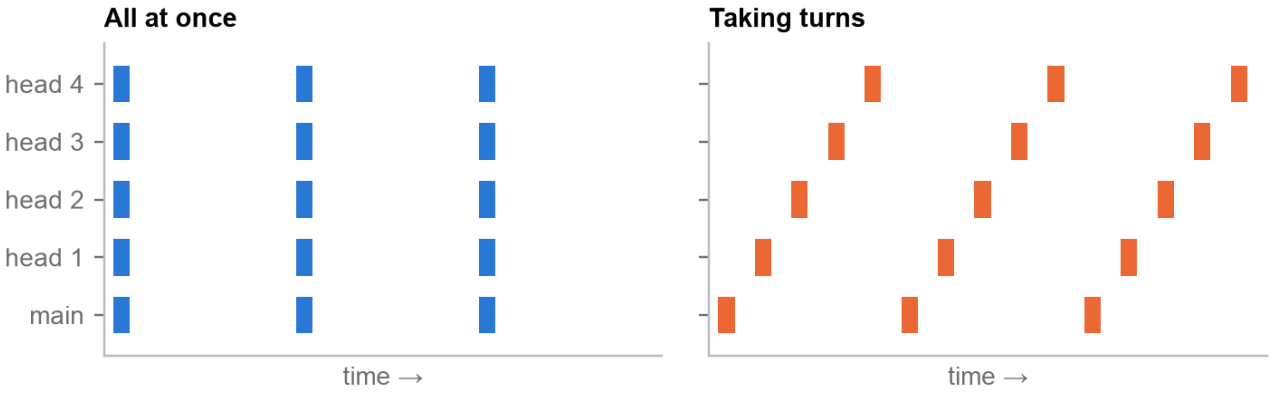
- Each coloured block is one voice's slice; the line in it is the part heard most.
- Head 2's arrow points left: it is in reverse.

14

How the voices share the work

Steps 2 and 3, listing the pitches and rebuilding the slice, are the hard work. Each voice does them once per half window, all in one go. At the largest window that takes over a millisecond per voice.

If every voice did its work at the same moment, the computer would face one big burst of work. VCV Rack has only a few milliseconds to deliver each small batch of audio, and a burst bigger than that causes a click or dropout. So the voices take turns: each one prepares its next slice a little early, at its own moment, and switches to it on time. The sound is exactly the same.



What to look for

- Left: all five voices work at the same moment, in one big burst.
- Right: the same amount of work, spread out so no two voices work at once.

Measured on the module with all 13 voices at the largest window: the busiest moment took about 7.2 milliseconds all at once, and about 0.6 milliseconds taking turns. Rack's limit for that batch is about 5.3 milliseconds.